Kinetic Sculpture Test Fixture Power Unit

Detailed Mechanical Design Document - Revision B

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Revision B focus: reduced heat-set insert use, press-fit fixed shaft mounts, unambiguous clamp/lug geometry, and revised repelling-magnet pawl bias.

Revision B Change Log

Area	Revision B change
Fastener philosophy	M3 heat-set inserts have been removed from the design. M3 screws either thread directly into printed pilot holes or use externally accessible captured M3 hex nuts where additional clamp force is useful. M4 heat-set inserts remain only at the standardized output interface, because that interface is expected to be reconfigured frequently.
Fixed shaft mounts	The fixed-shaft mount no longer uses a clamp slit or clamp screws. The 8 mm shaft is installed once into a tight blind bore and retained by press fit plus optional wicking CA adhesive.
Clamp slots and ears	All remaining split/clamp parts define whether the clamp ears are integral lugs or simply split body material. Each split slot, lug, screw axis, counterbore face, and threaded/pilot side is defined by local X/Y/Z coordinates.
Hidden insert pockets	No heat-set insert pockets are placed on internal slit faces or otherwise unreachable surfaces. All heat-set inserts are installed from accessible front faces only.
Freewheel pawl magnets	The previous small attraction magnets are replaced by larger 1/4 in x 1/8 in through-thickness neodymium magnets arranged as radial repulsion pairs. Each pawl has one magnet in its outer radial edge; the source pulley has a fixed magnet tower radially outside each pawl. Same poles face each other to push the pawl inward toward the external drum ratchet.
Part descriptions	Each fabricated part now includes a geometry narrative before the explicit CAD reconstruction list.

1. Executive Design Summary

This design defines a purely mechanical, wall-mounted power unit for a modular kinetic sculpture test fixture. It stores energy in a falling barbell-plate weight stack and delivers reasonably constant torque through a constant-radius cable drum, a printed freewheel ratchet, printed round-belt pulleys, and a standardized output hub. The weighted drum is a replaceable power-source module; the belt transmission and output hub can remain in place for future power-source variants such as a constant-force spring module.

The recommended baseline remains a 10 lb weight stack, 36 in of usable weight fall, a 70 mm cable-center drum pitch diameter, a 180 mm source pulley, and a 60 mm output pulley. With an assumed 80% transmission efficiency, baseline output torque is approximately 3.7 in-lbf (0.42 N-m), with about 12.5 output turns per wind. Source pulleys of 210 mm, 150 mm, and 120 mm pitch diameter provide discrete torque/runtime tuning without changing the weight stack.

Revision B intentionally simplifies fabrication. The shaft mounts are permanent press/adhesive mounts rather than adjustable split clamps. Serviceable but low-risk parts such as collars and pivots use direct-to-plastic pilot holes. The only heat-set inserts retained are the six M4 inserts in the output hub, where repeated adapter changes are expected.

2. High-Level Architecture and Layout

The system is divided into four modules: (1) weight and cable, (2) source drum and winding/freewheel module, (3) belt transmission, and (4) standardized output hub. Both rotating axes are perpendicular to a vertical plywood backer and use fixed 8 mm shafts with 608 bearings installed in the rotating printed parts. The falling weight is exposed and falls through open air below the source drum.

Recommended board layout, front view, 24 in x 24 in x 3/4 in plywood backer:

```
Y up
^
|     source/drum axis                output axis
|     X=6.00 in, Y=17.50 in           X=18.00 in, Y=17.50 in
|     O=========================================O     round metal belt
|     |                          |
|     | weight cable falls downward        | standardized hub to future escapement
|     v                          |
+-------------------------------------------------------------------> X right
Origin is lower-left front face of plywood. +Z points out from the wall toward the viewer.
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2.1 Mechanical Power Flow

- A coated stainless wire rope is clamped to the drum and wound into a single-layer helical groove. The weight stack pulls on the cable and applies nearly constant torque to the drum.
- During the power stroke, the drum rotates clockwise when viewed from the front. The drum rear external ratchet drives two source-pulley pawls, so the source pulley rotates with the drum.
- During winding, the detachable crank turns the drum counterclockwise viewed from the front. The drum ratchet slips under the source-pulley pawls, allowing the source pulley, belt, output hub, and downstream sculpture to remain stationary.
- The source pulley drives the output pulley with McMaster 6178K22 ultra-flexible round metal belting and 6178K32 thread-on hook connectors.
- The output pulley is integrated with a standardized six-bolt hub and a separate external anti-reverse ratchet. A gravity pawl blocks reverse output motion during winding.

2.2 Key Decisions, Constraints, and Assumptions

Requirement / constraint	Design response
Purely mechanical power unit	No motor, electronics, active brake, or powered clutch is included. Energy storage is a falling weight.
Reasonably constant torque	The drum uses a single-layer cable groove with a constant 35.0 mm cable-center radius. Constant weight force times constant radius gives nearly constant source torque.
Configurable torque	Primary adjustment is by source pulley pitch diameter. Secondary adjustment is weight-stack mass. No friction clutch is used for torque setting.
Winding at drum; output stationary	A source freewheel ratchet between drum and source pulley allows rewinding without driving the belt/output backward.
Anti-reverse protection	A separate output ratchet and gravity pawl blocks reverse output motion caused by freewheel drag during winding.

Requirement / constraint	Design response
Cantilevered fixed shafts	Both axes use 8 mm fixed shafts installed permanently into printed press-fit mount blocks. Rotating parts carry two 608 bearings each.
Minimal heat-set inserts	M4 heat-set inserts are used only on the repeatedly reconfigured output interface. All M3 locations use direct printed pilot holes or externally accessible captured nuts.
Bambu P2S / 256 mm build volume	All printed parts fit within 256 mm x 256 mm x 256 mm. The largest optional source pulley has 224 mm OD.
Projection from wall	Nominal maximum projection is about 130 mm (5.1 in) on the source axis and about 94 mm (3.7 in) on the output axis, excluding the removable winding crank.
No guards required	The falling weight, belt, and ratchets remain exposed, matching the input fixture requirement.

2.3 Board Coordinates and Axis Stack-Up

Layout item	Definition
Coordinate system	Origin at lower-left corner of plywood front face. +X right, +Y up, +Z outward from the wall.
Source axis	X = 152.4 mm (6.00 in), Y = 444.5 mm (17.50 in), Z axis normal to board.
Output axis	X = 457.2 mm (18.00 in), Y = 444.5 mm (17.50 in), Z axis normal to board.
Center distance	304.8 mm (12.00 in) between shaft centerlines.
Source-axis stack from board	Board; shaft mount body to Z about 42 mm; 1 mm thrust washer; 28 mm source pulley; about 1 mm controlled axial gap; 47 mm drum; 1 mm washer; 8 mm collar. Source shaft cut length: 130 mm.
Output-axis stack from board	Board; shaft mount body to Z about 42 mm; 1 mm thrust washer; 43 mm output pulley-hub; 1 mm washer; 8 mm collar. Output shaft cut length: 105 mm.
Weight path	Cable exits the drum near source-axis X and falls vertically downward. Keep the board clear below the source axis for 36 in of travel.

3. Calculations and Sizing Rationale

3.1 Energy and Drum Torque

The baseline weight is 10 lb and the usable fall distance is 36 in (914.4 mm). Stored energy is approximately 10 lbf x 36 in = 360 in-lbf = 40.7 J. The drum cable-center radius is 35.0 mm (1.378 in), so source drum torque at 10 lb is 13.8 in-lbf (1.56 N-m). The 36 in fall produces about 4.16 drum revolutions with a 70 mm cable-center pitch diameter.

3.2 Output Torque, Runtime, and Belt Length Variants

The output torque estimate assumes 80% overall mechanical efficiency from drum to output hub. The output pulley remains fixed at 60 mm pitch diameter. The source pulley is swapped to change speed-up ratio. Belt loop lengths below use the standard open-belt approximation for 304.8 mm center distance; final loops should still be cut by wrap-and-test because connector engagement and printed groove details affect installed length.

Mode	Source pulley pitch dia.	Ratio	Estimated output torque at 10 lb	Output turns per wind	Approx. belt loop length
Long-run / lowest torque	210 mm	3.50:1 speed-up	3.15 in-lbf (0.36 N-m)	14.6	1052 mm (41.4 in)
Baseline	180 mm	3.00:1 speed-up	3.67 in-lbf (0.42 N-m)	12.5	998 mm (39.3 in)
Medium-high torque	150 mm	2.50:1 speed-up	4.41 in-lbf (0.50 N-m)	10.4	946 mm (37.2 in)
Highest torque standard	120 mm	2.00:1 speed-up	5.51 in-lbf (0.62 N-m)	8.3	895 mm (35.2 in)

Torque scales linearly with weight. For example, the 120 mm source pulley gives about 0.62 N-m at 10 lb, about 0.75 N-m at 12 lb, and about 0.93 N-m at 15 lb before real-world friction differences. Runtime scales with output turns and inversely with average output speed. At the baseline 12.5 output turns per wind, an average escapement-controlled output speed of 0.2 rpm gives about 62 minutes, 0.5 rpm gives about 25 minutes, and 0.8 rpm gives about 16 minutes.

3.3 Estimated Target Output Torque Range

The probable test wheel is a 20 in diameter spoked plywood wheel with most mass near a 1-2 in wide outer ring, potentially carrying up to four 12 in satellite assemblies near the main wheel perimeter. A rough inertia envelope for such a fixture is approximately 0.08-0.18 kg-m^2, depending heavily on satellite mass and mounting radius. For slow escapement-driven operation, the power unit does not need to continuously accelerate this inertia at high speed; it mainly needs enough impulse energy to overcome friction, lift/drive mechanism elements, and advance the escapement in less than one full revolution.

This design therefore targets a practical output torque range of about 0.25-0.75 N-m (2.2-6.6 in-lbf), with the option to exceed the upper end during short tests by using the smaller source pulley and/or a heavier weight stack. The baseline 0.42 N-m mode is deliberately central; pulley swaps cover lower-torque/longer-runtime and higher-torque/shorter-runtime tests.

3.4 Torque Consistency

A falling weight has nearly constant force. The single-layer drum groove keeps the cable center radius at 35.0 mm, so the dominant source-torque variation is small compared with the target +/-10-15% tolerance. The largest remaining variation sources are printed bearing friction, pawl drag, belt slip/tension, cable routing angle, and weight stack friction if the plates rub during fall. The one-layer groove is a key design decision because it avoids the changing torque caused by winding cable over previous cable layers.

4. Sourced Parts Inventory - McMaster-Carr

Part-number schema: sourced parts use KSTF-SRC-###. McMaster-Carr catalog part numbers are listed separately. Quantities are for one fixture plus reasonable spares where noted. User-provided barbell plates are not included as McMaster-sourced fixture hardware.

Fixture sourced part	McMaster part	Name / description	Qty	Key dimensions and design use	Link
KSTF-SRC-001	6178K22	Ultra-flexible round metal belting, steel, 1/4 in OD, 10 ft length	1 roll	Belt OD 0.250 in (6.35 mm). Cut to length. Used on printed U-groove pulleys. Each pulley-ratio variant should have its own loop when practical.	mcmaster.com/6178K22
KSTF-SRC-002	6178K32	Thread-on hook connectors for 1/4 in ultra-flexible round belting, steel	1 pkg of 5	Connector threads onto each belt end and hooks together. Connector size matches 1/4 in belt.	mcmaster.com/6178K32
KSTF-SRC-003	5972K91	Open 608 ball bearing, steel	8 bearings	8 mm bore, 22 mm housing OD, 7 mm width, ABEC 1. Fabricated bearing pockets use 22.10 mm diameter x 7.20 mm depth. Six required, two spare.	mcmaster.com/5972K91
KSTF-SRC-004	4143N19	303 stainless steel rotary shaft, 8 mm diameter, 400 mm length	1	Diameter tolerance -0.005 to 0 mm. Cut into 130 mm source shaft and 105 mm output shaft. Shafts are fixed and do not transmit torque.	mcmaster.com/4143N19
KSTF-SRC-005	3459T42	Nylon-coated 18-8 stainless wire rope	1 spool	1/16 in rope diameter, 3/32 in coated diameter, 90 lb listed capacity, 7 x 7 strand core, 2-5/8 in recommended pulley/drum diameter. Used as falling-weight cable.	mcmaster.com/3459T42
KSTF-SRC-006	90591A121	M3 x 0.5 steel hex nuts	1 pkg	Nominal standard M3 hex nut: 5.5 mm across flats, about 2.4 mm high. Used only where a captured nut is explicitly specified; otherwise M3 screws thread into	mcmaster.com/90591A121

Fixture sourced part	McMaster part	Name / description	Qty	Key dimensions and design use	Link
				pilot holes.	
KSTF-SRC-007	94180A353	Brass heat-set insert for plastic, M4 x 0.7	1 pkg	Selected insert length 7.9 mm. Used only in KSTF-FAB-008 output interface. Front-access pockets are 5.3 mm diameter x 10.0 mm deep to tolerate modest substitute length variation.	mcmaster.com/94180A353
KSTF-SRC-008	91290A117	Socket head cap screw, alloy steel, M3 x 0.5 x 12 mm	1 pkg	Used for direct-thread pivot or light-retention screws, including output pawl pivot and crank knob. Typical head clearance used in printed parts: 6.2 mm dia x 3.3 mm deep.	mcmaster.com/91290A117
KSTF-SRC-009	91290A121	Socket head cap screw, alloy steel, M3 x 0.5 x 18 mm	1 pkg	Used where longer reach is needed: split collars, source freewheel pawl pivots, retainer ring clamp, and cable clamps.	mcmaster.com/91290A121
KSTF-SRC-010	91290A148	Socket head cap screw, alloy steel, M4 x 0.7 x 12 mm	1 pkg	Used for future attachments to the standardized output hub six-bolt interface.	mcmaster.com/91290A148
KSTF-SRC-011	5862K103	Neodymium disc magnet, 1/4 in OD x 1/8 in thick, N52, through-thickness magnetized	8 magnets	OD 6.35 mm, thickness 3.18 mm, listed pull about 2.8 lbf. Used in repelling radial pairs for source freewheel pawl bias. Install with same poles facing.	mcmaster.com/5862K103
KSTF-SRC-012	91070A203	Rounded-head wood screw for plywood/OSB, No. 8 x 1-1/2 in	1 pkg	Used to fasten printed mounts and pawl bracket to 3/4 in plywood. Printed bases are 22 mm thick; pilot-drill plywood.	mcmaster.com/91070A203
KSTF-SRC-013	75445A44	Thin instant-bond cyanoacrylate adhesive, 3M Scotch-Weld PR100	1 bottle optional	Optional wicking CA for permanent 8 mm shaft-to-mount retention after press installation. Thin liquid can wick into close seams; do not use on bearings or ratchets.	mcmaster.com/75445A44

4.1 Purchased Part Dimensional Interfaces Used in Fabricated Design

Purchased component	Nominal purchased dimensions	Fabricated interface definition
608 bearing	8.00 mm ID, 22.00 mm OD, 7.00 mm width.	Bearing pockets are 22.10 mm dia x 7.20 mm deep. Fixed shafts are 8 mm. Inner spacers and thrust washers use 8.20 mm ID so they slide freely.
8 mm shaft	8 mm diameter; selected shaft tolerance slightly under nominal (-0.005 to 0 mm).	Shaft-mount blind bores use CAD 8.00 mm diameter and are tuned by coupon or light reaming for light press fit. Optional CA is applied only after perpendicularity is verified.
1/4 in round metal belt	6.35 mm diameter.	Pulleys use a 7.2 mm wide U-groove with 3.6 mm groove radius. Pitch diameter is at belt centerline.
Wire rope	3/32 in coated OD = about 2.38 mm.	Drum groove is designed around 2.38 mm cable, 3.2 mm helix pitch, and 35.0 mm cable-center radius. Clamp channel is 3.2 mm wide.
M3 machine screws	M3 x 0.5. Head clearance based on about 5.5-5.7 mm head diameter and about 3 mm head height.	Clearance holes: 3.4 mm. Direct-thread pilot holes in PETG: 2.6 mm diameter unless noted. Counterbores: 6.2 mm diameter x 3.3 mm deep.
M3 hex nuts	5.5 mm across flats, about 2.4 mm high.	Captured-nut pockets, where used, are externally accessible and sized 5.7 mm AF x 2.7 mm depth, with a 3.4 mm screw clearance through the clamp ear.

Purchased component	Nominal purchased dimensions	Fabricated interface definition
M4 heat-set insert	M4 x 0.7, selected length 7.9 mm.	Only output hub uses heat-set inserts. Front-access pockets are 5.3 mm diameter x 10.0 mm deep.
Pawl magnets	1/4 in OD x 1/8 in thick = 6.35 mm x 3.18 mm, through-thickness magnetized.	Radial magnet pockets are 6.60 mm diameter x 3.35 mm deep. The pocket axis is radial, so magnet flat faces oppose each other across a radial air gap.

5. Fabricated Parts Inventory

Part-number schema: fabricated parts use KSTF-FAB-###. Swappable variants use letter suffixes. Unless explicitly noted, fabricated parts are 3D printed. PETG is preferred for loaded, ratcheting, shaft-mounting, and weight-support components. PLA is acceptable only for fit-checks or low-load templates.

Fabricated part	Name	Qty	Material	Primary function
KSTF-FAB-001	Standard fixed 8 mm cantilever shaft mount body	2	PETG	Permanent press/CA support for fixed shafts.
KSTF-FAB-002	8 mm printed split end-retainer collar	4	PETG	Prevents components sliding off shafts; sets axial position.
KSTF-FAB-003	Weighted drum spool with rear freewheel ratchet and winding sockets	1	PETG	Stores cable, receives crank, drives source pulley through ratchet.
KSTF-FAB-004A	Source drive pulley / freewheel pawl carrier, 180 mm pitch diameter	1	PETG	Baseline ratio.
KSTF-FAB-004B	Source drive pulley / freewheel pawl carrier, 210 mm pitch diameter	Optional 1	PETG	Longer runtime / lower torque.
KSTF-FAB-004C	Source drive pulley / freewheel pawl carrier, 150 mm pitch diameter	Optional 1	PETG	Medium-high torque.
KSTF-FAB-004D	Source drive pulley / freewheel pawl carrier, 120 mm pitch diameter	Optional 1	PETG	Highest standard torque.
KSTF-FAB-005	Freewheel pawl with radial repulsion magnet pocket	2 plus spare	PETG	Engages drum ratchet; slips during winding.
KSTF-FAB-006	Detachable four-pin winding crank arm	1	PETG	Manual winding interface at drum.
KSTF-FAB-007	Winding crank knob	1	PETG or PLA	Rotating grip for crank.
KSTF-FAB-008	Output pulley-hub with standard six-bolt interface and anti-reverse ratchet	1	PETG	Belt-driven output and future attachment hub.
KSTF-FAB-009	Gravity anti-reverse pawl	1 plus spare	PETG	Blocks reverse output rotation.
KSTF-FAB-010	Output anti-reverse pawl bracket	1	PETG	Holds gravity pawl at output ratchet.
KSTF-FAB-011	Barbell plate weight carrier	1	PETG	Carries user-provided plates and cable attachment.
KSTF-FAB-012	Weight carrier top retainer split ring with captured-nut lug	1	PETG	Keeps plates on carrier during handling.
KSTF-FAB-013	Cable clamp pressure plate	3	PETG	Clamps cable at drum and weight carrier; one spare.
KSTF-FAB-015A	Drum bearing inner-ring spacer sleeve	1	PETG	Prevents bearing preload; length 32.6 mm.
KSTF-FAB-015B	Source pulley bearing inner-ring spacer sleeve	1	PETG	Prevents bearing preload; length 13.6 mm.
KSTF-FAB-015C	Output hub bearing inner-ring spacer sleeve	1	PETG	Prevents bearing preload; length 25.6 mm.
KSTF-FAB-016	8 mm inner-race thrust washer	8+	PETG	1 mm axial spacers contacting bearing inner races only.

6. Common Fabricated-Part Design Rules

- Dimensional units: all fabricated part geometry is in millimeters unless noted otherwise. Board placement is also given in inches where useful.

- Coordinate convention for part specs: unless a part-specific origin is stated, local +Z is the part axis or the direction out from the board/front face. For flat pawls, local XY defines the outline and local Z is thickness.
- General noncritical printed dimensions: +/-0.25 mm. Functional bearing pockets, ratchet teeth, pilot holes, shaft bores, and magnet pockets: target +/-0.10 mm after printer calibration.
- Heat-set insert policy: use heat-set inserts only where frequent reassembly is expected. In this revision, only KSTF-FAB-008 uses M4 heat-set inserts. No M3 heat-set inserts are required anywhere in the fixture.
- Direct M3 pilot holes: use a 2.6 mm modeled pilot hole for M3 x 0.5 machine screws into PETG. If the printer creates undersized holes, clear them gently with a 2.5-2.6 mm drill by hand; do not enlarge to clearance size.
- M3 clearance holes: use 3.4 mm. M3 socket-head counterbores: 6.2 mm diameter x 3.3 mm deep unless a part-specific geometry says otherwise.
- Captured-nut pockets: only use captured nuts on externally reachable faces. Standard pocket is 5.7 mm across flats x 2.7 mm deep for an M3 hex nut. The nut must be insertable from an open outside face; no nut or insert pocket may be placed inside an inaccessible clamp slit.
- Clamp ears/lugs: when the word "ear" or "lug" is used, the spec explicitly defines it. A split slot alone does not imply extra rectangular ears. For parts with lugs, the lug X/Y/Z extents are listed.
- Bearing interface: use 22.10 mm diameter x 7.20 mm deep pockets for 608 bearings. Adjust only the bearing pocket diameter in 0.05 mm increments after fit-coupon testing.
- Round-belt grooves: pulley pitch diameter is at the belt centerline, not at printed rim OD. Use a 3.6 mm radius U-groove and enough flange height to retain a 6.35 mm belt.
- Pawl pivots: screws act as pivots, not clamps. Tighten until retained, then back off slightly or use a small thread-locking/CA dot on the screw head area only if needed. Do not glue the pawl to the pivot.
- Printed ratchet teeth must be clean and fully formed. Print ratchet wheels with the axis vertical so tooth profiles are resolved primarily in XY rather than layer stair-steps.
- Material: PETG is the default. PLA is acceptable for prototype fit checks but is more vulnerable to creep in shaft mounts, ratchets, collars, and weight-supporting parts.
- Print profile baseline: 0.20 mm layer height, 5-6 wall loops, at least 5 top/bottom layers, and 40-60% infill. Use near-solid walls or 100% infill for pawls, collars, washers, cable clamps, and small lugs.
- Shaft mount fit: print a simple 8 mm bore coupon before printing final shaft mounts. The final fixed-shaft bore should be a light press fit on the actual shaft. Use wicking CA only after shaft perpendicularity is checked.

6.1 Repelling Magnet Pawl Bias Rule

The source freewheel pawls use magnet repulsion, not attraction. Each pawl carries one 1/4 in x 1/8 in disc magnet in a cylindrical pocket cut radially into the pawl outer edge. Each source pulley has a fixed magnet tower radially outside each pawl with a matching radial magnet pocket. The disc magnets are through-thickness magnetized, so their flat faces must face one another across the radial air gap. Install the magnets so the same pole faces the same pole; this pushes the pawl inward toward the external drum ratchet. Mark one pole of every magnet with a permanent marker before installation, test the repulsion on the bench, and then glue the magnets into the pockets with CA.

7. Detailed Fabricated Component Specifications

KSTF-FAB-001 - Standard Fixed 8 mm Cantilever Shaft Mount Body

Purpose/function: Rigidly supports a fixed 8 mm shaft perpendicular to the plywood backer. The shaft is static; bearings inside rotating parts spin around it.

Geometry narrative: This part is a flat rectangular wall plate with a solid cylindrical boss growing outward from its center. The shaft enters the boss from the front and stops against a solid rear wall. There is no split, no clamp slit, no clamp ear, and no threaded insert. The geometry should look like a stout round socket fused to a rectangular base, with four board-screw holes around it. Once the shaft is pressed and optionally bonded into the bore, the mount is treated as permanent.

Detailed description and system fit: Two identical mounts are used, one at the source/drum axis and one at the output axis. Each mount screws to the plywood backer, aligns the 8 mm shaft along +Z, and provides a fixed cantilever support. Because the shaft never needs to be removed after installation, retention is by light press fit plus optional wicking CA, not a clamp.

Key interface requirements:

- Interfaces with KSTF-SRC-004 8 mm shaft, actual diameter 8.000 to 7.995 mm. Shaft insertion bore is blind and includes a 4 mm rear stop.
- Mounts to plywood using four KSTF-SRC-012 No. 8 x 1-1/2 in wood screws through counterbored holes.
- No heat-set inserts, no M3 clamp screws, and no unreachable pockets are present in this part.
- Local origin is at shaft centerline on the rear board-contact face. +X is horizontal on the board, +Y is vertical on the board, and +Z points outward from the board.

Full dimensional definition and feature reconstruction:

- Base plate: rectangular prism, X = -48 to +48 mm, Y = -41 to +41 mm, Z = 0 to 22 mm. Rear face at Z = 0 is flat and contacts the plywood.
- Cylindrical shaft boss: 42 mm OD cylinder centered on the shaft axis, X = 0, Y = 0, extending from Z = 0 to Z = 42 mm. Merge into base plate. Add 2 mm fillet at boss/base intersection.
- Blind shaft bore: 8.00 mm diameter CAD bore, coaxial with Z axis, cut from front face at Z = 42 mm back to Z = 4 mm. The solid wall from Z = 0 to 4 mm is the shaft insertion stop. The as-printed bore should be tuned by coupon to produce light press fit on the actual shaft.
- Bore entry: 0.8 mm x 45 degree chamfer at Z = 42 mm to start the shaft without shaving plastic.
- Board screw holes: four holes parallel to Z, 5.0 mm diameter through, at (X,Y) = (+/-34, +/-27). Add front counterbores 10.5 mm diameter x 6.0 mm deep from Z = 22 face.
- Optional shaft-depth witness window: do not cut through the boss. If a mark is desired, engrave a shallow 0.5 mm line on the outside of boss at Z = 4 mm to indicate rear-stop depth.
- All outside base edges have 1.5 mm fillets. Front boss rim has 0.8 mm chamfer. Board-contact rear face remains flat.

Print profile and design-for-manufacturing notes:

- Print rear face down on the bed in PETG, 0.20 mm layers, 6 walls, 50% infill or higher.
- Print a bore coupon first. If the shaft cannot enter with firm hand/arbor pressure, lightly ream or sand the bore. If the shaft slips freely, adjust CAD bore smaller for the final mounts or rely on CA only after confirming perpendicularity.
- Install the 130 mm source shaft and 105 mm output shaft until they contact the rear stop. Use a square to verify perpendicularity before applying optional KSTF-SRC-013 wicking CA at the bore entrance. Keep CA away from bearings.

KSTF-FAB-002 - 8 mm Printed Split End-Retainer Collar

Purpose/function: Prevents rotating assemblies and spacers from sliding off the fixed 8 mm shafts and provides axial adjustment during assembly.

Geometry narrative: This is a short round collar with a central shaft bore and one explicitly modeled side clamp. The main body is a cylinder. On the +X side, two rectangular lugs are merged into the cylinder, one above and one below a vertical slit. The slit cuts from the outside of the collar into the shaft bore. A screw crosses the slit from the +Y lug into a direct plastic pilot hole in the -Y lug. These lugs are separate rectangular features, not merely a round boss split by an unspecified slot.

Detailed description and system fit: A collar is placed near the front end of each fixed shaft after the rotating stack is installed. Additional collars may be printed for setup stops. The collar is not a high-force clamp; it only retains axial position and should not preload the bearings.

Key interface requirements:

- Interfaces with KSTF-SRC-004 8 mm fixed shaft. Bore is 8.10 mm for slip fit before tightening.
- Uses one KSTF-SRC-009 M3 x 18 screw threaded directly into a 2.6 mm pilot hole in the far lug. No heat-set insert is used.
- Should bear only against KSTF-FAB-016 thrust washers or bearing inner races/spacer sleeves, not bearing outer races or rotating plastic faces.
- Local origin is at collar center. Local Z is shaft axis. Local +X points toward the clamp lugs. Local +Y points toward the screw-head side.

Full dimensional definition and feature reconstruction:

- Main ring body: cylinder 18.0 mm OD x 8.0 mm long, centered on Z, with central 8.10 mm through bore.
- Clamp slit: rectangular slot centered on local +X radial direction, from X = +4.05 to +15.0 mm, Y = -1.0 to +1.0 mm, Z = 0 to 8.0 mm. The slot cuts from the bore to the outside through the +X side only.
- Clamp lugs: two rectangular lugs merged to the main cylinder on the +X side. Upper/head-side lug: X = +7.5 to +15.0, Y = +1.0 to +6.5, Z = 0 to 8.0. Lower/thread-side lug: X = +7.5 to +15.0, Y = -6.5 to -1.0, Z = 0 to 8.0.
- Clamp screw axis: line parallel to local Y at X = +11.5 mm, Z = 4.0 mm. From +Y outside face, cut a 3.4 mm clearance hole through the upper lug and across the 2.0 mm slit. Add a 6.2 mm diameter x 3.3 mm deep counterbore on the +Y face. In the lower lug, cut a coaxial 2.6 mm pilot hole to accept M3 threads. The pilot may pass through the -Y face.
- Chamfer both bore openings 0.5 mm. Chamfer screw-hole openings 0.3 mm. Fillet exterior lug corners 0.8 mm.

Print profile and design-for-manufacturing notes:

- Print in PETG with 100% infill or 6+ walls. Print with Z axis vertical so the bore is round, or on its side if that gives a cleaner clamp slit; verify the bore after printing.
- Tighten only enough to remove axial looseness. If the direct pilot strips, reprint the collar rather than converting it to a heat-set insert.
- Place a 1 mm KSTF-FAB-016 washer between collar and bearing inner race/spacer stack to avoid rubbing rotating outer races.

KSTF-FAB-003 - Weighted Drum Spool with Rear Freewheel Ratchet and Winding Sockets

Purpose/function: Stores the falling-weight cable on a constant-radius single-layer groove, receives the winding crank, and drives the source pulley through a printed freewheel ratchet during the power stroke.

Geometry narrative: This part is a 47 mm long rotating spool around the source shaft. The rear face carries an external ratchet ring. Moving forward, the ratchet is followed by a rear flange, a grooved cable barrel, a front flange, and a broad winding boss with four round sockets. The central hub contains one 608 bearing pocket from each side. The cable groove is a shallow helix around the barrel so the cable lays in one layer at a constant radius. The cable clamp is on the accessible front face and uses direct M3 pilot holes, not heat-set inserts.

Detailed description and system fit: The drum rotates on two 608 bearings around the source-axis fixed shaft. It sits in front of the source drive pulley. The rear external ratchet overlaps the two pawls mounted on the source pulley. The front face is exposed for winding-crank engagement and cable clamp service.

Key interface requirements:

- Uses two KSTF-SRC-003 608 bearings in 22.10 mm x 7.20 mm pockets and KSTF-FAB-015A inner spacer sleeve.
- Cable interface uses KSTF-SRC-005 coated wire rope with 2.38 mm coated OD. Drum groove pitch and clamp channels are sized from this value.
- Rear ratchet engages two KSTF-FAB-005 pawls on KSTF-FAB-004. Power direction is clockwise viewed from front. Winding direction is counterclockwise viewed from front.

- Front winding sockets accept four 5.8 mm pins on KSTF-FAB-006 winding crank.
- Cable clamp screws are KSTF-SRC-009 M3 x 18 into direct 2.6 mm pilot holes; no inserts are used.

Full dimensional definition and feature reconstruction:

- Coordinate: drum rear face is local Z = 0, front face is local Z = 47 mm. Axis is local Z. Through center bore is 8.6 mm diameter for shaft clearance.
- Central hub body: 32 mm OD cylinder, Z = 0 to 47 mm, merged with drum flanges and barrel.
- Bearing pockets: rear pocket from Z = 0 and front pocket from Z = 47, each 22.10 mm diameter x 7.20 mm deep. Include a 1.0 mm axial pocket-bottom lip at 20.0 mm diameter so the bearing seats on the outer-race region while leaving center clearance.
- Rear ratchet ring: external ratchet occupies Z = 0 to 7 mm. Root diameter 84 mm, tooth-tip diameter 92 mm, tooth axial width 7 mm.
- Ratchet tooth count: 64 teeth. Pitch angle p = 5.625 degrees. For tooth i in XY profile, use root radius Rr = 42.0 mm and tip radius Rt = 46.0 mm. Points are P0 at angle i*p, radius Rr; P1 at i*p + 0.75 degrees, radius Rt; P2 at i*p + 2.85 degrees, radius Rt; P3 at (i+1)*p, radius Rr. Connect P0-P1 as the near-radial drive face, P1-P2 as the tip land, P2-P3 as the freewheel ramp, and P3-next P0 as root arc. Add 0.4 mm root fillets.
- Rear flange: Z = 7 to 12 mm, 86 mm OD.
- Cable barrel: Z = 12 to 36 mm, base cylinder 67.6 mm OD. Add a helical rounded groove with cable-center radius 35.0 mm, groove profile radius 1.45 mm, and pitch 3.2 mm per revolution. Include 4.75 turns. Groove start is near Z = 13.0 mm at angular location 270 degrees. Groove end is near Z = 28.2 mm after 4.75 turns. The groove must carry only one cable layer.
- Front flange: Z = 36 to 44 mm, 86 mm OD.
- Front winding boss: Z = 44 to 47 mm, 68 mm OD. Four crank sockets are 6.3 mm diameter blind holes, 8.0 mm deep from the front face, on a 54.0 mm bolt circle at angles 45, 135, 225, and 315 degrees. Chamfer socket mouths 0.8 mm.
- Cable anchor: at groove start, route cable through a 3.2 mm tangential/radial hole into a straight clamp channel on the front face. Clamp channel is 3.2 mm wide x 2.8 mm deep x 24 mm long, centered on a chord outside the four crank sockets. Two 2.6 mm M3 pilot holes are centered 14 mm apart along the clamp-plate screw line, symmetric about the cable channel, and cut 10 mm deep from the front face. Pilot holes align with KSTF-FAB-013.

Print profile and design-for-manufacturing notes:

- Print with axis vertical in PETG. Use 0.20 mm layers, 6 walls, and 60% infill. Choose bed orientation that gives the cleanest ratchet teeth; usually rear face up or down should be tested on a small ratchet coupon first.
- Bearing pockets should accept bearings by firm hand pressure or light arbor pressure. Do not force bearings so hard the hub cracks.
- After installation, inspect ratchet teeth for strings, blobs, and rounded drive faces. Cable clamp pilot holes should accept M3 screws with firm thread forming, not loose clearance.

KSTF-FAB-004A/B/C/D - Source Drive Pulley and Freewheel Pawl Carrier

Purpose/function: Carries the round metal belt and the two freewheel pawls that couple the weighted drum to the belt transmission during the power stroke.

Geometry narrative: This part is a flat, bearing-supported round-belt pulley. Its front face also functions as the pawl carrier. Two pawl pivots are located on a 59 mm radius, opposite each other. Radially outside each pawl pivot is an integral rectangular magnet tower that protrudes from the same front face as the pawl. The magnet tower and the pawl magnet face each other along a radial line so magnet repulsion pushes the pawl inward. The pivot screws thread into direct pilot holes in printed bosses; no inserts are used.

Detailed description and system fit: The pulley rotates independently on two 608 bearings around the source fixed shaft. Its front face sits adjacent to the drum rear ratchet. Swappable pitch diameters set the torque/runtime ratio while keeping the output pulley fixed.

Key interface requirements:

- Uses two 608 bearings and KSTF-FAB-015B inner spacer sleeve.
- Uses KSTF-SRC-001 1/4 in round metal belt and KSTF-SRC-002 connector.
- Carries two KSTF-FAB-005 pawls on KSTF-SRC-009 M3 x 18 pivot screws threaded directly into printed pilot holes.
- Uses four KSTF-SRC-011 magnets total: one in each pawl and one in each fixed magnet tower. Magnets are radial, through-thickness, and installed same-pole facing for repulsion.
- Pitch diameter variants: A = 180 mm baseline, B = 210 mm long-run, C = 150 mm medium-high torque, D = 120 mm highest standard torque.

Full dimensional definition and feature reconstruction:

- Coordinate: rear face Z = 0, front pawl-mount face Z = 28 mm. Pulley axis is local Z. Through center bore is 8.6 mm diameter.
- Central hub: 36 mm OD cylinder, Z = 0 to 28 mm. Bearing pockets from both faces are 22.10 mm diameter x 7.20 mm deep with 20.0 mm diameter inner clearance lip as used on the drum.
- Rim and web: pulley body is 28 mm axial width. Web disk is at least 6 mm thick between hub and rim. Rim OD equals pitch diameter + 14 mm. Nominal rim ODs are: A 194 mm, B 224 mm, C 164 mm, D 134 mm.
- Round-belt groove: belt centerline is at radius Dp/2 and axial position Z = 14.0 mm. Cut a 7.2 mm wide U-groove with 3.6 mm groove radius. Groove lips have 0.8 mm rounds. Flange material must retain the belt at least 3 mm radially above the groove bottom.
- Pawl pivot local frames: two pivot centers at polar angles 90 degrees and 270 degrees, radius 59.0 mm. For each pivot, define local +Y as radial outward from pulley center, local -Y inward toward the ratchet, local +X tangent in the normal clockwise power direction, and local +Z outward from the source pulley front face.
- Pivot bosses: at each pivot, add a 14 mm OD x 2.0 mm high cylindrical boss from Z = 28.0 to 30.0 mm. Add a 2.6 mm diameter pilot hole centered on the pivot axis, depth 13.0 mm measured from Z = 30.0 mm. The M3 x 18 screw threads into this pilot after passing through the 7 mm pawl.
- Raised pivot washer pad: annular pad 10.5 mm OD x 0.6 mm high around each pivot, top at Z = 30.6 mm, so the pawl rear face clears the pulley face by 0.6 mm. Pawl rear face rides on this pad.
- Fixed magnet towers: for each pawl local frame, create a rectangular tower merged to the front face with extents local X = -7.0 to +7.0 mm, local Y = +9.5 to +18.5 mm, local Z = 28.0 to 36.5 mm. The tower inner face is the plane local Y = +9.5 mm, facing the pawl outer edge.
- Tower magnet pocket: in each tower, cut a 6.60 mm diameter x 3.35 mm deep cylindrical pocket with axis along local -Y. The pocket opens on the tower inner face at local Y = +9.5 mm, centered at local X = 0 and local Z = 32.25 mm. Magnet face should end flush with the inner face or recessed no more than 0.2 mm.
- Pawl swing clearance: cut/leave a flat area around each pivot so KSTF-FAB-005 can rotate at least 18 degrees outward and 8 degrees inward without rubbing the pulley face or magnet tower. The minimum air gap between pawl magnet face and tower magnet face in nominal engaged position is 2.0-2.5 mm.
- External fillets: 1.5 mm on rim edges, 1.0 mm on hub edges, 0.8 mm on tower outside edges. Bearing pocket entrances have 0.5 mm chamfer.

Print profile and design-for-manufacturing notes:

- Print in PETG with pulley axis vertical. For the 210 mm variant, use a brim and confirm the part remains flat; the 224 mm rim OD fits within the Bambu P2S bed.
- Use 5-6 walls and at least 40% infill. Make the pivot boss and magnet tower regions near solid with local modifier or high wall count.
- Install fixed tower magnets after confirming pawl swing. Mark magnet poles and use same-pole facing between tower and pawl to create repulsion. If pawl force is too high, increase the radial gap by sanding the tower inner face slightly or recessing one magnet deeper.
- Each pulley diameter should have its own belt loop or a length-adjusted loop. Do not rely on excessive belt stretch for ratio swaps.

KSTF-FAB-005 - Freewheel Pawl with Radial Repulsion Magnet Pocket

Purpose/function: Engages the weighted drum ratchet during the power stroke and slips over the ratchet ramps during winding.

Geometry narrative: This pawl is a small, flat, nearly triangular lever. Its pivot hole is near the outer end. The working nose extends radially inward toward the drum ratchet. The outer radial edge contains a horizontal/radial magnet pocket, not a face-on magnet pocket. The magnet flat face points radially outward toward the matching fixed magnet tower on the source pulley. Same-pole repulsion from the tower pushes the nose inward into the external ratchet teeth.

Detailed description and system fit: Two pawls are mounted on the source drive pulley front face. They are biased by magnet repulsion and pivot on M3 screws. Light but positive inward bias reduces winding friction while improving engagement reliability compared with the previous small attraction magnets.

Key interface requirements:

- Pivots on KSTF-SRC-009 M3 x 18 screw threaded into KSTF-FAB-004 pivot pilot hole. Pivot screw must not clamp the pawl tightly.
- Tip geometry interfaces with KSTF-FAB-003 64-tooth external ratchet, root radius 42 mm, tip radius 46 mm, 7 mm axial tooth width.
- Magnet pocket accepts KSTF-SRC-011 magnet, OD 6.35 mm x 3.18 mm thick. Pocket axis is local +Y and magnetization is along that axis after installation.
- Local frame: pivot hole center at (0,0). Local +Y points radially outward when installed. Local -Y points inward toward the ratchet. Local +X points tangent in the normal clockwise power direction. Part thickness is local Z.

Full dimensional definition and feature reconstruction:

- Part is a flat 7.0 mm thick plate. Define the 2D outline in local XY and extrude from Z = 0 to Z = 7.0 mm.
- Pivot hole: 3.3 mm diameter through hole centered at local (X,Y) = (0,0). Add 0.3 mm chamfer both sides.
- Main outline vertices before fillets, listed clockwise: A (-5.5,+7.0), B (+5.5,+7.0), C (+7.5,-6.0), D (+4.0,-13.2), E (+1.0,-16.2), F (-3.8,-15.0), G (-7.0,-6.0). Return to A. Fillet A, B, C, G with 1.0 mm. Keep nose features crisp except specified nose radius.
- Working nose: create a 0.6 mm radius nose centered near local (X,Y) = (+0.2,-16.0). The locking face is the line from approximate point (+1.0,-16.2) to (+4.0,-13.2); orient this face so drum load tends to hold the pawl inward rather than cam it outward. The opposite side from (-3.8,-15.0) to the nose is the freewheel ride face.
- Radial magnet pocket: cut a cylindrical pocket 6.60 mm diameter x 3.35 mm deep with axis along local +Y. Pocket opens on the outer edge plane Y = +7.0 mm. Pocket centerline is at local X = 0 and local Z = 3.5 mm. The bottom of the pocket ends at Y = +3.65 mm. Magnet should be flush or recessed 0.2 mm from outer edge.
- Add shallow embossed text or dot on the front face near the pivot: "OUT" at the +Y side to avoid installing the pawl backward. Embossing height/depth 0.4 mm maximum.

Print profile and design-for-manufacturing notes:

- Print flat in PETG, 100% infill or enough perimeters to make the part effectively solid. Use 0.12-0.20 mm layer height.
- Clean the pivot hole and working nose. The pawl must swing freely on an M3 screw with no side binding.
- Before gluing magnets, mark magnet poles. Install the pawl magnet so it repels the tower magnet when the pawl is mounted. If the pawl is pushed outward instead of inward, remove and reverse one magnet before final glue cure.
- If winding friction is too high, increase magnet gap, recess one magnet slightly, or remove one pawl temporarily for diagnostic testing. Final operation should use two pawls.

KSTF-FAB-006 - Detachable Four-Pin Winding Crank Arm

Purpose/function: Provides a removable hand crank that engages the drum front sockets for winding. The crank is not left installed during normal operation.

Geometry narrative: The crank has a round hub with four short rear-facing pins that plug into the drum sockets. A single rounded arm extends from the hub to a handle boss. The handle boss has a direct M3 pilot hole for the knob screw; no heat-set insert is used. The crank is a removable tool, so its extra wall projection does not count against normal mounted projection.

Detailed description and system fit: The crank is pressed onto the drum front face so four printed pins enter four blind sockets. Turning counterclockwise viewed from the front winds the cable back onto the drum while the drum freewheels relative to the source pulley.

Key interface requirements:

- Four pins interface with 6.3 mm sockets in KSTF-FAB-003. Pins are 5.8 mm OD for clearance.
- Central clearance hole avoids the fixed shaft and front collar.
- Handle boss accepts KSTF-FAB-007 knob using KSTF-SRC-008 M3 x 12 screw threaded into a direct printed pilot hole.

Full dimensional definition and feature reconstruction:

- Coordinate: hub center is local origin. Hub disk lies in XY, crank thickness along Z. Rear face is Z = 0, front face is Z = 12 mm. Rear pins protrude in negative Z from Z = 0 to Z = -8 mm.
- Hub disk: 78 mm OD x 12 mm thick. Central clearance hole 18 mm diameter through.
- Rear engagement pins: four pins, 5.8 mm OD x 8.0 mm long, centered on 54.0 mm bolt circle at angular positions 45, 135, 225, and 315 degrees. Pin ends have 0.5 mm chamfers.
- Crank arm: rounded rectangle from hub center to handle boss center, 120 mm center distance along local +X. Arm is 18 mm wide in Y x 12 mm thick in Z and tangent/merged to hub disk. Add 3 mm end fillets and 3 mm hub-arm transition fillets.
- Handle boss: 24 mm OD x 12 mm thick cylinder centered at local X = +120 mm, Y = 0, Z = 0 to 12. Add 2.6 mm diameter pilot hole 10 mm deep from front face Z = 12, centered on boss axis, for M3 knob screw.
- Optional direction marking: engrave "WIND CCW" and an arrow on the front face at 0.5 mm depth.

Print profile and design-for-manufacturing notes:

- Print front face on bed with rear pins upward, or rear face on bed if pin quality is better with supports. PETG preferred, 5 walls, 50% infill.
- During winding, push the crank fully into the drum sockets before applying torque. Remove crank after winding.

KSTF-FAB-007 - Winding Crank Knob

Purpose/function: Provides a comfortable rotating hand grip for the detachable crank.

Geometry narrative: The knob is a simple rounded cylinder with an M3 clearance hole along its axis. The screw head is recessed at the outer end. The knob rotates around the screw shank; it is not a precision bearing. No insert is used in the knob.

Detailed description and system fit: The knob mounts to the handle boss on KSTF-FAB-006 with an M3 screw. It only needs to spin freely enough for hand winding.

Key interface requirements:

- Central clearance hole for KSTF-SRC-008 M3 x 12 screw.
- Outer shape should be comfortable and free of sharp edges.
- Interfaces with crank boss pilot hole in KSTF-FAB-006; screw should be left slightly loose so knob spins.

Full dimensional definition and feature reconstruction:

- Coordinate: knob axis is local Z. Rear face toward crank is Z = 0, outer face is Z = 32 mm.
- Knob body: cylinder 28 mm OD x 32 mm long.
- Central through hole: 3.4 mm diameter along Z. Outer-face counterbore at Z = 32 face: 6.5 mm diameter x 4.0 mm deep for M3 screw head.
- Round both outside end edges with 2.0 mm fillets.

- Optional grip grooves: eight equally spaced shallow axial grooves, 1.0 mm deep, along the outside surface. Grooves are nonfunctional and may be omitted.

Print profile and design-for-manufacturing notes:

- Print in PETG or PLA. 3-4 walls and 25% infill are sufficient.
- Do not tighten the knob screw until the knob binds. Leave slight axial clearance so it can rotate.

KSTF-FAB-008 - Output Pulley-Hub with Standard Interface and Anti-Reverse Ratchet

Purpose/function: Receives belt power, provides the standardized fixture output hub, and carries an anti-reverse ratchet to protect downstream mechanisms during winding.

Geometry narrative: This part is the main output rotating component. The rear/outer region contains the anti-reverse ratchet teeth. Around the middle is the fixed 60 mm pitch round-belt groove. The front face contains a raised 50 mm pilot and a six-bolt M4 insert pattern for future adapters. This is the only part in Revision B that uses heat-set inserts, because the output interface is expected to be reconfigured repeatedly. All insert pockets are on the front face and are fully reachable with a soldering-iron tip.

Detailed description and system fit: The output pulley-hub rotates on two 608 bearings around the output-axis fixed shaft. Future escapements, pulleys, gears, or test-wheel adapters attach to the M4 six-bolt pattern and raised pilot.

Key interface requirements:

- Uses two 608 bearings and KSTF-FAB-015C inner spacer sleeve.
- Belt groove interfaces with KSTF-SRC-001 1/4 in round metal belt.
- Six KSTF-SRC-007 M4 heat-set inserts on a 60.0 mm bolt circle accept KSTF-SRC-010 M4 screws for future attachments.
- Raised front pilot is 50.0 mm OD for concentric registration of future adapters.
- Rear/outer ratchet interfaces with KSTF-FAB-009 gravity anti-reverse pawl.

Full dimensional definition and feature reconstruction:

- Coordinate: rear face Z = 0, standard interface front face Z = 40 mm, raised pilot extends Z = 40 to 43 mm. Axis is local Z. Through bore is 8.6 mm diameter.
- Main body: 80 mm OD cylinder, Z = 0 to 40 mm. Bearing pockets: 22.10 mm diameter x 7.20 mm deep from both faces with standard 20.0 mm inner clearance lip.
- Output belt pulley: fixed pitch diameter 60.0 mm. Groove centerline radius is 30.0 mm, axial location Z = 18.0 mm. Cut a 7.2 mm wide U-groove with 3.6 mm radius. Rim/flange OD around pulley zone is 74 mm minimum.
- Anti-reverse ratchet ring: external ratchet occupies Z = 3 to 10 mm. Root diameter 86 mm, tooth-tip diameter 94 mm, tooth axial width 7 mm.
- Anti-reverse ratchet tooth count: 72 teeth. Pitch angle p = 5.0 degrees. Use root radius 43.0 mm and tip radius 47.0 mm. For tooth i, points are P0 at i*p, root radius; P1 at i*p + 2.15 degrees, tip radius; P2 at i*p + 3.00 degrees, tip radius; P3 at (i+1)*p, root radius. Orient this ratchet to allow clockwise output rotation and block counterclockwise reverse rotation.
- Standard output pilot: 50.0 mm OD x 3.0 mm high, centered on axis, on front face. Add 0.5 mm chamfer at pilot top edge.
- Attachment pattern: six M4 heat-set insert pockets on 60.0 mm bolt circle at angles 0, 60, 120, 180, 240, and 300 degrees. Each pocket is 5.3 mm diameter x 10.0 mm deep from the front face. Pockets are blind and accessible from the front. No pocket is on a slit or hidden surface.
- Center front clearance: 18 mm diameter recess/clearance through the pilot area so the fixed shaft and retaining collar do not interfere with future attachments.
- Optional label: emboss or engrave "KSTF-60" and a clockwise arrow on the front face at 0.6 mm depth/height outside the 50 mm pilot but inside the 80 mm OD.

Print profile and design-for-manufacturing notes:

- Print in PETG with axis vertical, rear face on bed. Use 6 walls and 50-60% infill around the output interface.

- Install M4 heat-set inserts from the front only. Keep inserts square and do not distort the raised pilot surface.
- After assembly, verify clockwise output rotation is free and counterclockwise rotation is blocked by the gravity pawl. Lift pawl manually only for service/reset.

KSTF-FAB-009 - Gravity Anti-Reverse Pawl

Purpose/function: Blocks counterclockwise reverse rotation of the output hub while allowing normal clockwise power rotation.

Geometry narrative: This pawl is a narrow flat lever that hangs from a pivot above the output ratchet. Its weight biases the lower toe into the ratchet. The geometry is deliberately simple and contains no magnet pocket, no spring pocket, and no insert. The pivot hole is plain clearance for an M3 screw.

Detailed description and system fit: The pawl mounts to KSTF-FAB-010 above the output ratchet. It is biased by gravity only, so engagement force is light and winding drag is low. The pawl can be manually lifted for service.

Key interface requirements:

- Pivots on KSTF-SRC-008 M3 x 12 screw threaded into KSTF-FAB-010 direct pilot hole. Screw must retain but not clamp the pawl.
- Tip engages KSTF-FAB-008 external ratchet at approximately 47 mm radius.
- No spring or magnet required. Bracket location sets engagement depth.

Full dimensional definition and feature reconstruction:

- Part is a flat 5.0 mm thick plate. Define local 2D geometry in XY and extrude Z = 0 to 5.0 mm.
- Pivot hole: 3.3 mm diameter through hole at local (0,0).
- Outline vertices before fillets: (-5,+5), (+5,+5), (+6,-25), (+3,-34), (-3,-34), (-6,-25). Fillet non-tip corners 1.0 mm.
- Working toe: at local Y = -34 mm, use 0.6 mm nose radius and a 10 degree locking-face back angle. Nose should sit into the output ratchet root without bottoming hard.
- Add 0.3 mm chamfers to pivot hole and non-working edges.

Print profile and design-for-manufacturing notes:

- Print flat in PETG at 100% infill or near-solid walls. Keep the nose crisp.
- During setup, place bracket so the pawl drops reliably into ratchet but rides over teeth during normal clockwise rotation with minimal clicking force.

KSTF-FAB-010 - Output Anti-Reverse Pawl Bracket

Purpose/function: Holds the gravity anti-reverse pawl at the correct radius and axial position relative to the output ratchet.

Geometry narrative: The bracket is a wall-mounted pedestal with a broad rear base and a forward cantilever arm. The front end of the arm contains a short round pivot boss. The pawl pivot screw threads directly into a pilot hole from the front face; there is no insert. The bracket is installed after the output hub is mounted so its position can be adjusted to the printed ratchet.

Detailed description and system fit: The bracket mounts directly to the plywood backer above the output axis. Because printed ratchet and pawl fit can vary slightly, the bracket is positioned by test fit, then pilot-drilled into the plywood.

Key interface requirements:

- Mounts to plywood with four KSTF-SRC-012 wood screws.
- Uses one KSTF-SRC-008 M3 x 12 screw as the pawl pivot threaded into a direct 2.6 mm pilot hole.
- Local coordinates are defined relative to the output shaft center: bracket base is above the output axis. Local +Z points outward from board.

Full dimensional definition and feature reconstruction:

- Base plate: rectangular prism X = -45 to +45 mm, Y = +38 to +94 mm, Z = 0 to 22 mm. Rear face Z = 0 contacts board.
- Board screw holes: four 5.0 mm through holes parallel to Z at X = +/-30 mm and Y = +50 and +82 mm. Add front counterbores 10.5 mm diameter x 6.0 mm deep.
- Cantilever arm/web: central rectangular prism X = -14 to +14 mm, Y = +38 to +72 mm, Z = 22 to 50 mm. Merge into base with 2 mm fillets.
- Side gussets: add triangular gussets on both sides, each 6 mm thick in X. Gussets span from Y = +42, Z = 22 to Y = +70, Z = 50.
- Pivot boss: centered at X = 0, Y = +66 mm, front face at Z = 50 mm. Boss is 16 mm OD x 5 mm protrusion, extending Z = 45 to 50 mm. Direct pivot pilot hole is 2.6 mm diameter x 10 mm deep along -Z from front face Z = 50. Add 0.5 mm chamfer.
- Pawl clearance relief: provide a flat pad or shallow 0.5 mm recess around pivot so KSTF-FAB-009 can swing without rubbing the bracket body.

Print profile and design-for-manufacturing notes:

- Print rear face down in PETG with 6 walls and 50% infill. Support should not be needed if gussets are modeled cleanly.
- Install bracket after output hub installation. Hold pawl in place, rotate output clockwise, adjust bracket so clockwise is free and counterclockwise is blocked, then drill pilot holes for wood screws.

KSTF-FAB-011 - Barbell Plate Weight Carrier

Purpose/function: Carries the user-provided Olympic-style plate stack and attaches the falling cable to the mass stack.

Geometry narrative: The carrier is a vertical printed spindle with a wide bottom support flange, a cylindrical center post sized to pass through 2 in Olympic plate holes, and a top head containing the cable clamp. The cable clamp is on the top face and uses direct M3 pilot holes, not inserts. Plates slide down over the spindle and rest on the bottom flange; the retainer ring is installed above the plates.

Detailed description and system fit: The carrier is not guided by rails and falls through open air. The cable clamps at the top using KSTF-FAB-013. The design assumes small barbell plates approximately 3.94 in OD with approximately 2 in center holes.

Key interface requirements:

- User plates assumed approximately 3.94 in OD and approximately 2 in center hole. The post is 46.0 mm OD to clear a 50.8 mm nominal hole.
- Cable clamp uses KSTF-SRC-005 wire rope, KSTF-FAB-013 pressure plate, and two KSTF-SRC-009 M3 x 18 screws into direct pilot holes.
- Recommended working load: 8-12 lb. Maximum recommended test load before revalidation: 15 lb.

Full dimensional definition and feature reconstruction:

- Coordinate: vertical spindle axis is local Z. Bottom support starts at Z = 0.
- Bottom flange: 82 mm OD disk x 8 mm thick, Z = 0 to 8 mm. Round outside lower edge with 2 mm fillet.
- Center spindle: 46.0 mm OD cylinder, Z = 8 to 125 mm. Add 5 mm fillet where spindle meets bottom flange.
- Top head: 58 mm OD disk, Z = 125 to 137 mm. Round upper edge 1.5 mm.
- Cable channel on top head: straight 3.2 mm wide x 2.8 mm deep x 28 mm long channel across top face. Channel is centered on local X axis from X = -14 to +14, Y = 0, cut downward from Z = 137.
- Cable pass hole: 3.2 mm diameter vertical hole from top face into channel at X = 0, Y = 0.
- Cable clamp pilot holes: two 2.6 mm diameter M3 pilot holes, 10 mm deep from top face, centered at X = +/-7.0 mm, Y = +6.0 mm relative to channel centerline so they match KSTF-FAB-013 holes when the clamp plate bridges the channel. If using a centered plate layout, keep 14 mm spacing and ensure screws do not intersect the cable hole.
- Optional plate-height marks may be embossed on spindle at 10 mm intervals; these are nonfunctional.

Print profile and design-for-manufacturing notes:

- Print upright in PETG with 6+ walls and 60-80% infill. Increase infill or wall overlap around spindle/flange transition and top cable clamp pilots.
- Before use, test over a soft surface with at least 2x planned operating weight. Inspect for layer splitting near flange and cable clamp.
- Because load is suspended, inspect cable clamp and printed spindle before each session. This is a test fixture component, not an overhead lifting component.

KSTF-FAB-012 - Weight Carrier Top Retainer Split Ring with Captured-Nut Lug

Purpose/function: Keeps plates on the weight carrier during handling and prevents plates from sliding off the top of the spindle if the carrier is tilted.

Geometry narrative: This is a large split ring that slides over the 46 mm weight-carrier spindle. On the +X side it has two explicit rectangular clamp lugs separated by a radial slit. Unlike the small shaft collars, this ring uses a captured M3 hex nut in the far lug because it may be opened and closed more often while changing weight plates. The nut pocket is on an outside face and is fully reachable.

Detailed description and system fit: The retainer sits above the weight stack on KSTF-FAB-011. It is not intended to carry the suspended load; it only prevents loose plates from lifting or sliding during handling.

Key interface requirements:

- Interfaces with KSTF-FAB-011 spindle OD 46.0 mm. Ring ID is 46.5 mm for slip fit before clamping.
- Uses one KSTF-SRC-009 M3 x 18 screw and one KSTF-SRC-006 M3 hex nut in an externally accessible captured-nut pocket.
- Local origin at ring center. Local Z is vertical spindle axis. Local +X points toward clamp lugs. Local +Y points toward screw-head side.

Full dimensional definition and feature reconstruction:

- Ring body: cylinder 72 mm OD, 46.5 mm ID, 10 mm thick, centered on local Z, Z = 0 to 10.
- Radial split slot: rectangular slot centered on +X radial side, from X = +23.25 to +46.0 mm, Y = -1.25 to +1.25 mm, Z = 0 to 10. Slot cuts from ID to outside through the +X side only.
- Clamp lugs: upper/head-side lug X = +33 to +46 mm, Y = +1.25 to +9.0 mm, Z = 0 to 10. Lower/nut-side lug X = +33 to +46 mm, Y = -9.0 to -1.25 mm, Z = 0 to 10. These are explicit rectangular features merged to the ring OD.
- Clamp screw axis: line parallel to local Y at X = +40.0 mm, Z = 5.0 mm. From +Y outside face, cut 3.4 mm clearance through upper lug, slit, and lower lug. Add 6.2 mm diameter x 3.3 mm head counterbore on +Y face.
- Captured nut pocket: on -Y outside face of lower lug, centered on the screw axis, cut a hexagonal pocket 5.7 mm across flats x 2.7 mm deep along +Y. Pocket axis is the screw axis. Pocket is open to -Y face so the M3 nut can be pushed in from outside. The 3.4 mm clearance hole passes through the pocket center.
- Chamfer top and bottom ID/OD edges 0.5 mm. Fillet exterior lug corners 1.0 mm.

Print profile and design-for-manufacturing notes:

- Print flat in PETG with 100% infill or 6+ walls. Verify the nut can be inserted from the outside face before use.
- Clamp lightly; the ring only restrains plates during handling and should not crush the printed spindle. If the nut pocket loosens, reprint the ring.

KSTF-FAB-013 - Cable Clamp Pressure Plate

Purpose/function: Clamps the coated wire rope into printed channels at the drum and weight carrier without metal crimp sleeves.

Geometry narrative: This is a small rectangular bridge plate with two screw clearance holes and a shallow underside groove. It presses the coated cable into a matching channel in either the drum front face or the weight carrier top head. The plate itself has no threads, no inserts, and no nuts.

Detailed description and system fit: The same clamp plate is used at the drum cable anchor and the weight carrier cable anchor. One spare is specified. The clamp is serviceable and re-adjustable, but the mating screws thread into direct pilot holes in the target printed part.

Key interface requirements:

- Uses two KSTF-SRC-009 M3 x 18 screws through 3.4 mm clearance holes. Screws thread into 2.6 mm pilot holes in KSTF-FAB-003 or KSTF-FAB-011.
- Underside groove interfaces with KSTF-SRC-005 3/32 in coated wire rope.
- Screw-hole pattern must match the drum and weight carrier clamp pilot spacing: 14 mm between hole centers.

Full dimensional definition and feature reconstruction:

- Plate coordinate: length along local X, width along local Y, thickness along local Z. Top face Z = +4, underside Z = 0.
- Plate: rectangular prism X = -12 to +12 mm, Y = -6 to +6 mm, Z = 0 to +4 mm.
- Screw holes: two 3.4 mm clearance holes through Z, centered at X = +/-7.0 mm, Y = 0.
- Underside cable groove: centered along local X at Y = 0, running full length X = -12 to +12. Use a rounded or V-groove 3.2 mm wide x 1.25 mm deep cut upward from underside Z = 0.
- Top edges chamfer 0.5 mm. Underside edges adjacent to cable groove should be smooth, not sharp enough to cut cable coating.

Print profile and design-for-manufacturing notes:

- Print flat in PETG with 100% infill. Ensure screw holes are clean and underside groove is smooth.
- When clamping cable, tighten both screws gradually and evenly. Tug-test over a soft surface before adding full weight. Re-tighten after the first few cycles if the cable coating compresses.

KSTF-FAB-015A/B/C - Bearing Inner-Ring Spacer Sleeves

Purpose/function: Prevents axial preload of paired 608 bearings by spacing the bearing inner races apart inside each rotating component.

Geometry narrative: Each sleeve is a simple printed tube. It slides over the fixed 8 mm shaft and sits between the two 608 bearing inner races inside a rotating component. When the front collar is adjusted, axial force travels through the collar, thrust washer, bearing inner race, spacer sleeve, opposite inner race, and shaft mount, instead of crushing the printed hub.

Detailed description and system fit: Each rotating component uses two 608 bearings. The inner spacer sleeve fits around the fixed shaft between bearing inner races. It should not touch bearing seals/shields or outer races.

Key interface requirements:

- Interfaces with KSTF-SRC-004 8 mm fixed shaft and KSTF-SRC-003 608 bearing inner races.
- Sleeve OD is small enough to contact only the bearing inner race, not the ball cage or outer race.
- All variants share 12.0 mm OD and 8.20 mm ID. Length varies by component body thickness.

Full dimensional definition and feature reconstruction:

- All variants: simple tube, 12.0 mm OD, 8.20 mm ID, square ends, 0.3 mm chamfers at both ends.
- KSTF-FAB-015A drum sleeve length: 32.6 mm. This matches a 47.0 mm drum body with two 7.20 mm bearing pocket depths.
- KSTF-FAB-015B source pulley sleeve length: 13.6 mm. This matches a 28.0 mm source pulley body with two 7.20 mm bearing pocket depths.
- KSTF-FAB-015C output hub sleeve length: 25.6 mm. This matches a 40.0 mm output hub body with two 7.20 mm bearing pocket depths.
- Length tolerance should be +/-0.10 mm where practical. If slightly long, sand ends square until bearings seat fully.

Print profile and design-for-manufacturing notes:

- Print vertically in PETG with 100% infill or high wall count. Use a brim if needed for stability.

- Confirm each sleeve slides freely over the 8 mm shaft before installing bearings. Remove elephant-foot material at sleeve ends.

KSTF-FAB-016 - 8 mm Inner-Race Thrust Washer

Purpose/function: Provides axial spacing where a rotating component, mount, or collar could otherwise rub on a bearing outer race or printed face.

Geometry narrative: This is a thin annular washer sized to touch only the inner race/spacer stack of a 608 bearing. Its OD is intentionally smaller than the 608 outer race, and its ID clears the fixed 8 mm shaft. Multiple washers are used as sacrificial printed shims during assembly.

Detailed description and system fit: Washers are placed between the shaft mount and first rotating component, between adjacent rotating components when needed, and between the front bearing inner race and the retaining collar. They contact only the bearing inner race/spacer stack.

Key interface requirements:

- Interfaces with KSTF-SRC-004 8 mm fixed shaft and KSTF-SRC-003 608 bearing inner race.
- OD must be smaller than bearing outer race and printed bearing-pocket features.
- Use one or more washers to set axial clearance without preloading bearings.

Full dimensional definition and feature reconstruction:

- Washer body: 12.0 mm OD, 8.20 mm ID, 1.0 mm nominal thickness.
- Print at least eight washers. Optional thickness variants of 0.5 mm and 1.5 mm may be printed for axial tuning; if used, mark thickness on part or storage bag.
- Chamfer edges lightly, 0.2-0.3 mm. Faces should be flat and free of strings.

Print profile and design-for-manufacturing notes:

- Print flat in PETG with 100% infill. Clean the ID before assembly.
- Set final axial play so rotating parts do not wobble excessively but still spin freely. Slight perceptible axial clearance is better than preloaded bearings.

8. Shaft and Bearing Stack Assembly Definitions

Axis	Fixed shaft preparation	Component stack-up
Source shaft	Cut 130 mm from KSTF-SRC-004. Deburr both ends. Insert into KSTF-FAB-001 until it reaches the 4 mm rear stop. Verify perpendicularity, then optionally wick KSTF-SRC-013 CA into the mount bore entrance.	From board outward: mount, 1 mm thrust washer, source pulley with spacer sleeve and bearings, about 1 mm axial gap/ratchet overlap adjustment, drum with spacer sleeve and bearings, 1 mm thrust washer, front retaining collar.
Output shaft	Cut 105 mm from KSTF-SRC-004. Deburr both ends. Insert into second KSTF-FAB-001 until rear stop. Verify perpendicularity, then optionally wick CA at mount bore entrance.	From board outward: mount, 1 mm thrust washer, output pulley-hub with spacer sleeve and bearings, 1 mm thrust washer, front retaining collar.

9. Assembly Procedure

1. Print all baseline fabricated components: two shaft mounts, four collars, weighted drum, 180 mm source pulley, two freewheel pawls plus spare, output pulley-hub, output pawl and bracket, weight carrier, top retainer, cable clamps, spacer sleeves, and thrust washers.
2. Print fit coupons for the 8 mm shaft bore, 608 bearing pocket, M3 direct pilot hole, and 1/4 in magnet pocket. Adjust only local feature sizes if necessary before printing the full part set.
3. Install only the six M4 heat-set inserts in KSTF-FAB-008. Do not install heat-set inserts in M3 holes; M3 locations are direct pilot holes or captured-nut pockets as specified.
4. Cut the 8 mm shaft stock into a 130 mm source shaft and 105 mm output shaft. Deburr and polish cut ends so bearings and collars slide on without shaving plastic.

5. Mount the two KSTF-FAB-001 shaft blocks on the plywood at the specified coordinates. Use pilot holes for the #8 wood screws. Verify each shaft is perpendicular to the board using a square before optional CA bonding.
6. Press 608 bearings into the source pulley, drum, and output hub. Place the correct KSTF-FAB-015 spacer sleeve between bearing inner races before the second bearing is fully seated.
7. Install source pulley on source shaft with a 1 mm thrust washer against the shaft mount. Install the two freewheel pawls and magnets on the pulley front face. Pawls must swing freely and be pushed inward by magnet repulsion.
8. Install drum in front of source pulley so its rear ratchet overlaps source-pulley pawls. Set a small axial clearance so pawls engage ratchet teeth but rotating bodies do not rub.
9. Install thrust washer and split collar at front of source shaft. Tighten collar only enough to retain the stack while preserving free rotation.
10. Install output pulley-hub on output shaft with thrust washers and front collar. Rotate clockwise by hand to confirm free movement.
11. Hold anti-reverse pawl bracket above output ratchet. Install gravity pawl on pivot, adjust bracket position so pawl allows clockwise rotation and blocks counterclockwise rotation, then pilot-drill and screw bracket to board.
12. Attach cable to drum clamp. Wrap cable into helical groove in a single layer, route it downward, and attach other end to weight carrier top clamp. Keep cable vertical under source drum as much as practical.
13. Cut and assemble the round metal belt for selected pulley ratio. Start with belt length estimate, wrap around pulleys, trim gradually, and install threaded connector. Belt should be taut enough to transmit load but not so tight bearings bind.
14. Load weight carrier with small test mass first. Verify power rotation, source freewheel engagement, output anti-reverse action, winding isolation, and cable retention. Increase weight gradually to 10 lb baseline.

10. Operation and Tuning

10.1 Normal Power Operation

- Lift or wind the weight carrier to the top of the allowed 36 in travel, then remove the crank.
- The weight pulls the cable, the drum rotates clockwise viewed from the front, the freewheel pawls drive the source pulley, and the belt drives the output hub clockwise.
- A downstream escapement should meter output motion. The power unit itself is not a speed governor.

10.2 Winding

- Insert the four crank pins fully into the drum sockets.
- Turn the crank counterclockwise viewed from the front. The drum rewinds the cable while its rear ratchet slips under the source-pulley pawls.
- The output hub should remain stationary. If belt/output tries to reverse due to freewheel drag, the gravity anti-reverse pawl blocks that motion.
- Remove the crank after winding. Power delivery during winding is not provided or required.

10.3 Torque and Runtime Tuning

- Use the 210 mm source pulley when long runtime and lower impulse torque are more important than maximum torque.
- Use the 180 mm source pulley as the baseline starting point for most escapement testing.
- Use the 150 mm or 120 mm source pulleys when higher output torque is required. Expect fewer output turns per wind.

- Change the weight stack only as a secondary adjustment. The recommended first adjustment is pulley ratio, because reducing weight also reduces stored energy and runtime.
- If belt slip occurs at high torque, slightly shorten the belt loop, improve pulley surface finish, or reduce acceleration load at the escapement. Avoid excessive belt tension that bends shafts or overloads bearings.

11. Verification Tests and Acceptance Criteria

Test	Acceptance criterion / method
Shaft perpendicularity	With shafts installed and before CA bonding, place a square against the board and shaft. Shaft should be visibly perpendicular. Correct by remounting before glue cures.
Shaft retention	After optional CA cure, try hand-pulling and twisting the shaft. It should not slip in the mount under hand force. Do not use the mount if the bore is loose without adhesive.
Bearing fit	Each rotating part should spin freely on the fixed shaft with no scraping. Bearing outer races should be retained by the printed pocket without cracking the hub.
Source pawl magnet bias	With drum removed, each freewheel pawl should be pushed inward by magnet repulsion. Moving the pawl outward should increase repulsion. If a pawl is pushed outward, reverse one magnet.
Source freewheel function	With belt removed, rotate drum clockwise: source pulley should follow. Rotate drum counterclockwise: source pulley should remain mostly stationary while pawls slip over ratchet ramps.
Output anti-reverse	Rotate output hub clockwise: it should move freely. Attempt counterclockwise rotation: gravity pawl should block within one or two ratchet teeth.
Winding isolation	Install belt and a light weight. Wind the drum with crank. Output hub should not rotate backward.
Cable retention	Clamp cable at drum and weight carrier, load with 2-3 lb over soft surface, tug/settle, and retighten. Increase to 10 lb only after no slip is observed.
Baseline performance	With 10 lb weight and 180 mm source pulley, confirm the system can turn output through about 12 rotations over the full 36 in fall under low load.

12. Safety, Durability, and Service Notes

- This is an exposed test fixture. Keep fingers, clothing, and loose tools away from belt, ratchets, and falling weight path.
- The selected wire rope is used as a fixture cable, not as an overhead lifting system. Do not stand under the suspended mass and do not use this design to lift people or valuable objects.
- The printed weight carrier and cable clamps must be inspected before each test session. Replace parts that show cracking, layer separation, ratchet tooth rounding, or cable coating damage.
- Do not exceed 15 lb on the weight carrier without revalidating shaft deflection, printed mount strength, wood screw pullout, cable clamps, and ratchet tooth load.
- If the fixture is left loaded, printed ratchet and mount components may creep over time, especially in warm rooms. Unload the weight after testing.
- No guard is included because the input requirements did not require one. A future guard can be added without changing powertrain geometry.

13. Future Constant-Force Spring Compatibility

The design intentionally separates the weighted-drum power-source module from the belt/output module. A future constant-force spring source can replace the drum/weight carrier if it drives either the same source drive pulley or a compatible printed pulley on the source axis. The output hub, anti-reverse pawl, belt selection, fixed shaft mounts, and board coordinate system do not depend on the falling-weight source. A future module should preserve the same power rotation direction and should either include its own winding isolation or continue to use a freewheel interface at the source pulley.

14. Open Items for CAD Release and First Prototype

- Printer-specific fit coupons should be made for 608 bearing pockets, 8 mm shaft bores, M3 direct pilot holes, and 1/4 in magnet pockets before printing the full pulley/drum set.
- Actual barbell plate center-hole diameter should be measured. If the hole is less than 48 mm, reduce KSTF-FAB-011 spindle OD or sand the spindle for clearance.
- Actual belt loop lengths should be finalized by wrap-and-trim assembly because round-belt connectors and printed pulley groove depth affect final installed length.
- Escapement torque and impulse behavior should be measured after the first prototype; source pulley variants can then be prioritized around actual load data.
- Magnet force should be tested with the printed gap. The design defines the geometry, but final pawl feel can be tuned by magnet recess depth, shim tape, or slight tower sanding.

15. Source Notes for Purchased Components

The McMaster-Carr part details used in this design were checked against product pages where available. Catalog availability, packaging, and product pages can change, so the listed dimensions and fabricated interface definitions are the controlling engineering inputs. Verify the catalog page before ordering.

Part numbers	Source detail used
6178K22 / 6178K32	McMaster ultra-flexible round belting: coiled-metal construction can be cut to length; 1/4 in x 10 ft steel belting is 6178K22 and 1/4 in connector package is 6178K32.
5972K91	McMaster 608 open steel bearing listing: 8 mm shaft diameter, 22 mm housing ID/OD interface, 7 mm width, ABEC 1.
4143N19	McMaster 8 mm 303 stainless rotary shaft listing: 400 mm length and diameter tolerance -0.005 to 0 mm.
3459T42	McMaster coated 18-8 stainless wire rope listing: 1/16 in rope, 3/32 in coated OD, 90 lb capacity, 7 x 7 strand core, 2-5/8 in recommended pulley/drum diameter.
94180A353	McMaster M4 x 0.7 heat-set insert listing: 7.9 mm length. Retained only for output interface.
5862K103	McMaster N52 neodymium disc magnet listing: 1/4 in OD, 1/8 in thick, through-thickness magnetized, about 2.8 lbf pull.
75445A44	McMaster 3M Scotch-Weld PR100 instant-bond adhesive listing: thin cyanoacrylate suitable for metal/plastic/rubber; used optionally for fixed shaft retention.

Appendix A - CAD Coordinate Summary

This appendix consolidates core numerical geometry. The detailed part sections above remain the controlling specifications.

Feature	CAD value
Board source axis	X = 152.4 mm, Y = 444.5 mm, Z outward
Board output axis	X = 457.2 mm, Y = 444.5 mm, Z outward
Fixed shaft diameter	8.00 mm purchased shaft; mount bore CAD 8.00 mm tuned by coupon; collars 8.10 mm; spacers/washers 8.20 mm ID
608 bearing pocket	22.10 mm dia x 7.20 mm deep
Drum pitch geometry	70.0 mm cable-center pitch diameter; 35.0 mm radius; 4.75-turn helical groove; 3.2 mm pitch
Drum ratchet	64 teeth; root dia 84 mm; tip dia 92 mm; tooth axial width 7 mm
Source pulley variants	Dp = 210, 180, 150, 120 mm; rim OD = Dp + 14 mm; groove for 6.35 mm belt
Source pawl pivots	Radius 59.0 mm; angles 90 and 270 deg; M3 x 18 into 2.6 mm pilot holes
Source pawl magnets	6.60 mm dia x 3.35 mm radial pockets; 2.0-2.5 mm nominal face-to-face gap; same-pole repulsion
Output pulley	Dp = 60.0 mm; groove for 6.35 mm belt; integrated output hub
Output interface	Raised pilot 50.0 mm OD x 3.0 mm high; six M4 heat-set inserts on 60.0 mm BCD
Output anti-reverse ratchet	72 teeth; root dia 86 mm; tip dia 94 mm; tooth axial width 7 mm
Weight carrier spindle	46.0 mm OD spindle; 82 mm OD bottom flange; 58 mm OD top head
Cable clamp plate	24 x 12 x 4 mm; two 3.4 mm holes spaced 14 mm; underside groove 3.2 mm wide x 1.25 mm deep